6

Vital rates and population dynamics

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6.1 Introduction

Population dynamics is the study of changes in the number of individuals in populations over time. Chapter 3 describes methods for estimating the abundance of marine mammal populations. Superficial characterization of population dynamics can be obtained by making a series of population estimates over time. Marine mammal ecologists and conservation biologists, however, are typically interested in the underlying issues regarding the mechanics and drivers of trends in population size. The change in abundance during a given period is, quite simply, the sum of additions to the population (births and immigrants) minus losses from the population (deaths and emigrants). Setting aside immigration and emigration for now, we will focus on the *vital rates*—reproductive rate and survival.

Countless environmental factors (including, but by no means limited to, climate–ocean variability, changes in prey, levels of predation, fisheries by-catch, disease, competition, and habitat alteration) affect the chances that individual animals will survival or give birth in a given year. Vital rates are the aggregate manifestation of these individual-level responses to the environment summed over all animals in a population. Marine mammals are large, long-lived vertebrates and their vital rates vary with age and sex. Therefore, we usually attempt to ascertain sex- and age-specific, or at least stage-specific (i.e. calf/pup, immature, adult) vital rates. Moreover, information on temporal and spatial variability in vital rates is critical for understanding both the intrinsic (age-structure, density dependence) and extrinsic environmental drivers of variable vital rates. The latter is a rapidly advancing area of interdisciplinary research where links are being established between marine mammal demography, climate–ocean variability, foraging behaviour, and health parameters. Characterizing these connections is critical at this time when anthropogenic alteration of marine mammal habitats, through, for example, global climate change, are predicted to escalate (e.g. Ragen *et al.* 2008).

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Characterization of vital rates is most informative when integrated with other information (such as population age/sex structure) in an appropriately structured demographic model. Even when little data exist, a model can be used heuristically to identify critical data gaps. Given at least some demographic data, modelling may allow a more holistic approach to diagnosing a decline. For example, Holmes *et al.* (2007) inferred temporal changes in both reproductive rates and survival based upon trends in the counts and age structure of Steller sea lions (*Eumetopias jubatus*). A model populated with more comprehensive vital rates estimates affords the opportunity to predict future trends and evaluate management options. Taking this approach to analyse the critically endangered North Atlantic right whale population, Fujiwara and Caswell (2001) concluded that if the deaths of just two adult females could be avoided each year, the population would cease to decline. In this chapter we present methods for estimating survival and reproductive rates in marine mammals, and provide an overview of approaches to marine mammal population modelling.

6.2 Reproductive rate

Reproduction can be studied using either longitudinal or cross-sectional approaches, both of which require large sample sizes. Longitudinal studies involve collecting information on individual reproductive histories, ideally throughout their entire lives. Compared to cetaceans, pinniped populations have been more extensively studied using longitudinal approaches because individuals can more readily be live-captured for marking (Lunn and Boyd 1991; Laws 1993; Le Boeuf and Laws 1994; Bowen *et al.* 2006). Longitudinal studies of most cetacean populations have relied on re-sighting histories of uniquely marked individuals (Hammond *et al.* 1990; Barlow and Clapham 1997; Mann 2000, Hamilton *et al.* 2007), although temporary capture to identify uniquely marked individuals has also been used (Wells *et al.* 1987; Wells 1991). Cross-sectional studies rely on observations collected once from many individuals in the population. Although cross-sectional data can be used to calculate a variety of reproductive parameters, they cannot be used to address individual annual variation or examine lifetime reproductive output. Reproductive parameters from many cetacean populations have been determined using cross-sectional data obtained from directed takes (Marsh and Kasuya 1984) and fisheries by-catch (Read *et al.* 2006). Pinniped populations have also been studied using this approach (Bowen *et al.* 1981; Stirling 2005).

6.2.1 Description of parameters

Fecundity

Fecundity is usually defined as the average number of female offspring born per female per unit time in either the population (gross) or in a given female age class (age-specific) (Gotelli 2001). However, the term fecundity is sometimes used differently

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depending upon the context. For example, in population models dealing with males and females, fecundity may be used to refer to total offspring per female.[1] Fecundity should not be confused with reproductive rate, which can be defined using natality (the proportion of females giving birth in a defined period), pregnancy rate (the proportion of pregnant females in a sample), or ovulation rate (the proportion of a sample females showing evidence of ovulation) (Perrin and Reilly 1984; Huber *et al.* 1991). Reproductive rates can also be reported for different age groups (Lunn *et al.* 1994). Next we consider the components of fecundity, i.e. pregnancy and birth rate.

Pregnancy and birth rate

The reproductive cycle of females can be divided into seven distinct events: ovulation, conception, implantation, gestation, birth, lactation, and weaning (Boyd *et al.* 1999a). Rates can be calculated for any of these stages but, because of stage-specific failures, these rates will differ. Although all are important from a life history perspective, the most important to the present discussion are pregnancy and birth rates.

Gross pregnancy rates (GPR) combine ovulation and implantation stages, and measure the proportion of pregnant females in a sample of the population (Read 1990; Laws, R. M. and Sinha 1993):

$$\text{GPR} = \text{number of pregnant females/number of sexually mature females}$$

Gross birth rate (GBR), or natality, relates specifically to parturition (process of giving birth) and specifies the proportion of sexually mature females that actually give birth (Croxall and Hiby 1983):

$$\text{GBR} = \text{number of parturient females/number of sexually mature females}$$

Unlike fecundity, birth rate quantifies neonates of both sexes. Although the birth rate will ultimately be used in most demographic analyses, determining the pregnancy rate is also important because significant differences in these parameters can reveal that there may be environmental constraints on reproduction. Age-specific pregnancy and birth rates can also be estimated by calculating the individual rates for each age class i (Croxall and Hiby 1983; Huber *et al.* 1991):

$$\text{BR}_i = \text{number of parturient females at age } i\text{/number of females at age } i$$

Age-specific analysis describes the lifetime pattern of reproduction, which is required for matrix population models (see Section 6.4.2), whereas a gross analysis only provides the average birth rate for the population. Age-specific data are also

[1] The fecundity parameter in Leslie matrix population models may be a combination of birth rate and survival of females or neonates (Caswell 1991).

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especially useful for comparing reproductive rates among populations where age structure differs (Harting *et al.* 2007).

Although we have referred mainly to age-specific analyses in this chapter, we only use age as an example of any state variable that could be used to subdivide a population into logical units. Caswell (2001) demonstrates stage-based approaches to population modelling. Age-based approaches are one special case of stage-based approaches. An alternative to using age is to use an indicator of body size. If energy-based approaches to understanding population dynamics are to be used then it may be more productive to structure population models by size rather than age, or at least to use both as state-based vectors.

Age at sexual maturity

This varies among species, as well as among and within populations. Variation in age at sexual maturity (ASM) influences reproductive output both for the individual and the population (Stearns 1992). Age at sexual maturity may also be density-dependent (Bowen *et al.* 1981; Read and Gaskin 1990). Therefore, ASM should not be viewed as a static parameter, especially in exploited populations or those living in unstable environments (Roff 1992).

For females, a sexually mature individual is defined as one which has ovulated at least once and that may have a *corpus luteum* or *corpus albicans* (see 'Ovarian and uterine analysis' in Section 6.2.2) in either ovary [see Harrison (1969) for a description of cetacean reproductive anatomy and ovulation cycle and Boyd *et al.* (1999a) for pinnipeds and sirenians]. For males, sexual maturity has been less precisely defined, but generally is assumed to be an individual having evidence of spermatogenesis in the central portion of the testis and that may have sperm present in the epididymis (Perrin and Reilly 1984; Laws, R.M. and Sinha 1993). These definitions may not always hold in species, such as many pinnipeds, that have highly seasonal reproductive cycles. Male reproduction is usually not considered in most demographic analyses and population models. ASM and age at first reproduction may differ, with the latter referring to the age at which primiparity (first birth event for a female) occurs.

Age at sexual maturity can be estimated from data obtained by longitudinal or cross-sectional sampling (Bowen *et al.* 2006; Hadley *et al.* 2006; Westgate and Read 2006; Danil and Chivers 2007). There are several analytical approaches to determine ASM in populations. The first, called the sum of fraction immature algorithm, is a non-parametric technique that was developed by DeMaster (1978) and subsequently modified by Hohn (1989). ASM is estimated as the sum of the fraction of immature individuals in each indeterminate age class (age classes in which both immature and mature individuals occur) added to the age of the first indeterminate age class. Following Hohn (1989):

$$ASM = j + \sum_{i=j}^{k} p_i x_i$$

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where j is the age of the youngest mature animal, k is the age of the oldest immature animal, p_i is the proportion of immature animals in age class i, and x_i is the number of age classes combined in age class i to achieve a sample size of at least 2. The variance can be estimated using:

$$Variance\ (ASM) = \frac{\sum p_i q_i x_i}{N_i - 1}$$

where q_i is the proportion of mature individuals in age class i and N_i is the sample size for age class i.

Another common technique is to estimate when 50% of the population has matured. This is typically done using logistic regression analysis (Caswell *et al.* 1998; Danil and Chivers 2007). Confidence intervals can be constructed using bootstrap techniques (Caswell *et al.* 1998; Danil and Chivers 2007).

An important distinction between ASM and age at physical maturity (APM) is that the latter is defined as the age at which an individual reaches its asymptotic size (either standard length or body mass). There can be a considerable lag between ASM and APM in some marine mammal species, especially amongst males (e.g. Olesiuk *et al.* 1990a; Read and Hohn 1995).

Senescence

Reproductive senescence describes a reduction in fecundity at older ages (Promislow 1991) and is poorly understood in most marine mammal species. This is partly because these long-lived vertebrates have not been studied sufficiently long to collect the necessary data, or when they have, the sample sizes in the oldest age classes are too small to be informative. Also compounding the difficulties in documenting senescence is the possibility that females with lower lifetime productivity may live longer, thereby falsely indicating senescence. Still, evidence of senescence has been reported in grey seals (Bowen *et al.* 2006), Hawaiian monk seals (Harting *et al.* 2007), subantarctic fur seals (Lunn *et al.* 1994), and northern elephant seals (Sydeman *et al.* 1991). It has also been inferred in cross-sectional studies on odontocetes by documenting decreases in the pregnancy rate in increasingly older age classes (see Perrin *et al.* 1976, 1977). It is difficult to detect senescence because relatively few females live to the advanced ages when senescence is expressed. This also means it will usually have little impact on population dynamics so, from this narrow perspective, it may not be a particularly important demographic feature.

6.2.2 Techniques

Sighting histories of marked individuals

Long-term studies based on re-sighting marked individuals have provided the most comprehensive estimates of reproductive parameters (Olesiuk *et al.* 1990a;

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Sydeman *et al.* 1991; Barlow and Clapham 1997; Pomeroy *et al.* 1999; Mann 2000; Wells 2003; Kendall *et al.* 2004; Bowen *et al.* 2006; Hadley *et al.* 2006; Harting *et al.* 2007). Ideally, marked animals will be of known age, either because they were first tagged (see Chapter 2) as neonates or by ageing individuals using extracted teeth (Wells 1991; Arnbom *et al.* 1992).

Documenting reproduction is often straightforward; observing a female giving birth or nursing a neonate is sufficient to confirm parturition. However, lack of such observations does not necessarily indicate a female did not reproduce. To obtain reliable estimates of natality, for example, field protocols should be designed to maximize the probability of detecting births. This works well in colonial pinnipeds with highly seasonal parturition. Alternatively, a way must be devised to estimate the probability of re-sighting reproductive and non-reproductive females (e.g. capture–recapture methods which are discussed in Section 6.3 on survival estimation).

Census-based studies

In any situation where the abundance of sexually mature females and young of the year can be estimated, it is possible to obtain an estimate of gross (as opposed to age-specific) reproductive rates (e.g. Johanos *et al.* 1994). This approach has been applied at seal colonies where marked mothers are repeatedly documented with marked offspring. It has also been used in free-swimming cetacean populations where calves are identified based on their swimming position (Perryman *et al.* 2002). The latter application can give biased results if mothers and calves are not associating as predicted, or if there is social segregation of the population where the presence of calves co-varies with the underlying social process.

Determination of pregnancy

Marine mammals have streamlined bodies which can make it difficult or impossible to determine if a female is pregnant by visual inspection, therefore both direct and indirect methods are used to determine pregnancy status.

Ovarian and uterine analysis

The most reliable way to determine pregnancy is to conduct a direct macroscopic examination of the ovaries and uterus after they are removed during a routine necropsy (but see the Ultrasound section, p. 126) (Perrin and Donovan 1984; Laws, R. M. and Sinha 1993). Each uterine horn is dissected longitudinally along its mid-line, starting from the cervix. This should reveal the presence of an embryo or fetus. In both pinnipeds and odontocetes, the early stages of pregnancy are difficult to observe with the naked eye, so the absence of a fetus does not rule out that the individual was pregnant. The dimensions of the uterine horns increase during pregnancy and can therefore be used to support the assignment of a reproductive status to an individual. It is also important to examine the ovaries. These are

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examined grossly for the presence of a corpus luteum that indicates pregnancy (Perrin and Donovan 1984). As pregnancy progresses the corpus luteum enlarges, approaching the mass of the rest of the ovary. When a corpus luteum regresses there is extensive fibrosis and hyalinization (Harrison *et al.* 1972) and these scars, called corpora albicantia, are believed to remain for life in some species (some cetaceans). Counts of corpora albicantia from individuals may be used in some circumstances to calculate reproductive histories, but it is important to document the life-history (e.g. Boyd 1984) of corpora lutea and corpora albicantia before using this ovarian anatomy to derive information about reproductive rates.

Hormone analysis

The gonadal steroid hormones, mainly progesterone and oestradiol, may be used for determining pregnancy. Both blood and blubber samples (St. Aubin 2001) have been used, but much care is required with the assays for these hormones because of the potential for cross-reaction with irrelevant compounds in the tissue samples. Considerable effort is required to validate the assays being carried out. Remote blubber biopsies can be obtained from wild individuals for hormone analysis (Kellar *et al.* 2006), which means that, assuming assays can be appropriately validated, this is a promising technique for sampling large numbers of individuals with minimal intervention (Krützen *et al.* 2002). Pregnant animals tend to have elevated levels of hormones (Mansour *et al.* (2002), relative to non-pregnant or immature animals, but covariates are sometimes required to fully discriminate pregnant from non-pregnant individuals (Gardiner *et al.* 1996). This approach has several limitations. For example, hormone levels may not be significantly elevated during the early stages of pregnancy, resulting in a false-negative diagnosis; and there may be overlap between the normal range of hormone concentrations in pregnant and non-pregnant individuals. Consequently, individual variation can make establishing a threshold level to indicated pregnancy difficult, and therefore species-specific validation studies are necessary before wild populations can be assessed. Reproductive status may also be determined by quantifying hormone metabolites in faecal samples (Rolland *et al.* 2005). Hormones levels are typically quantified using standard radio- and enzyme-immunoassay techniques.

In general, hormone determinations of pregnancy based upon gonadal steroids do present significant challenges, and are probably most useful when sampling from individuals sequentially to examine the relative level of these hormones. Interestingly, in common with humans but with very few other mammals, pinnipeds (and possibly also cetaceans) produce a chorionic gonadotrophin (Hobson and Boyd 1984). This means that it may be possible to test for pregnancy using a specific test like the human pregnancy test. However, the protein structure of the gonadotrophins in marine mammals is sufficiently different from the human to make human pregnancy test kits ineffective for marine mammals.

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Ultrasound

Ultrasonography (diagnostic B-mode) has been used on both seals (Adam *et al.* 2007) and dolphins (Stone 1990; Brook *et al.* 2001) to visualize the presence of a developing fetus as well as follicular and corpus luteum development. Ultrasound can be conducted on cetaceans directly, but seals require shaving (fur compromises image quality) or transrectal and transvaginal approaches (Adams *et al.* 2007). While this technique has not been widely used in field situations, it is commonly used in captive and husbandry facilities (Brook *et al.* 2001). Ultrasound imaging is becoming more accessible to researchers with the development of portable diagnostic units (Adam *et al.* 2007). Its main limitations are that it requires trained expertise to interpret the sonograms and the animal must be captured.

6.3 Survival rate

As with reproductive parameters, survival can be estimated from cross-sectional and longitudinal observations. The former involves inferring survival from a sample of the population age structure (i.e. number or proportion of individuals in a population by age), whereas the latter involves observing individually marked animals over time. Because marine mammals are long-lived, we are usually interested in *annual* survival rates, rather than shorter intervals. Likewise, population models that incorporate survival rates typically operate on an annual time increment so that survival from one year to the next is the parameter of interest. However, in principle, methods using marked individuals can be used to evaluate survival over any time interval desired.

6.3.1 Cross-sectional age structure analysis

A traditional method for estimating the survival of wildlife populations involves acquiring a sample of animals and evaluating their age composition at a given point in time. The sample might come from a directed research effort, but it could also derive from commercial, subsistence, or recreational hunting.

Age structure analysis infers survival rates by examining the number of individuals in adjacent age classes. For example, a sample with $50 \times$ 5-year-olds and $40 \times$ 6-year-olds would yield an estimated 80% survival rate from age 5 to 6 (that is $40/50 = 0.8$). Calculating this ratio for every age available results in a set of age-specific rates. While intuitive and simple, age structure approaches are fraught with problems related to strong and rarely valid assumptions. The first issue is that the age structure of the sample must be representative of the age composition of the population of interest, which is equivalent to assuming that all individuals in the population have the same probability of being selected. In cases of commercial or recreational hunting, it is unlikely that animals taken will be representative of the total population, either due to hunter selectivity or differences in the accessibility of different age groups (Roff and Bowen 1986). However, even if the sample were

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representative, another issue remains. For an observed pattern of dwindling numbers with increasing age to truly reflect survival rates, additional strong assumptions are required. First, the population must have a stable age distribution, meaning that the proportion of animals in each age group does not change from year to year. Also, total population abundance and survival rates must be constant over time. Why are these conditions required? Consider what happens, in the example above, if survival were to vary. Our estimate of survival from age 5 to 6 years assumed that the proportion of 5-year-olds in the sample we obtained remained constant from the previous year (when our current 6-year-olds were 5-year-olds). That is, had we taken an equal number of animals the previous year, we would have also counted $50 \times$ 5-year-olds. However, what if survival was particularly poor in the past year and there had actually been $80 \times$ 5-year-olds? Then the real survival rate would have been 50% ($40/80 = 0.5$) rather than the 80% we estimated. Likewise, let's consider what happens if reproduction (and consequently abundance) fluctuated. If there is a bumper crop of offspring one year followed by dismal production the next, one might find a case where there are more yearlings than neonates. That would lead to an impossible estimate of a survival greater than 100%.

It is rarely the case that any wildlife population has a stable age distribution and stationary abundance. Rather, environmental conditions vary, and predator and prey populations fluctuate, which buffets vital rates and keeps age structures dynamic. In fact, it is often precisely because we suspect a change in vital rates has led to a population decline that we are interested in estimating these parameters. If that is the case then using age structure to estimate survival is perilous indeed.

While less than ideal, cross-sectional age structure methods may yield the best available estimates of survival. Realities of funding and logistics often dictate that optimal approaches to estimating vital rates cannot be achieved. If teeth from hunted seals or from dolphins incidentally caught in a fishery are available, these should be used to their fullest potential (e.g. Ferguson *et al.* 2005). A thorough treatment of age structure approaches to estimating survival is presented in Skalski *et al.* (2005).

6.3.2 Capture–recapture

Approaches to estimating survival using marked animals include 'known fate' models (where all individuals marked can be found again and consequently their fate is known), so-called 'band recovery' models where marked animals are re-identified only when they are dead (usually associated with hunting/harvesting), and finally 'live recapture' models whereby, as the name implies, live animals are re-sighted or recaptured. Models which combine both live and dead encounters also exist. Here, we will focus primarily on live recapture, or Cormack-Jolly-Seber (CJS) survival estimation (Cormack 1964; Jolly 1965; and Seber 1965) and its broader application (Lebreton *et al.* 1992), with some discussion of other model types.

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Studies that involve observing individually identifiable animals over time are variously referred to as 'mark–recapture', 'capture–recapture', or 'capture–mark–recapture' and additional information about these models applied to estimating abundance is given in Chapter 3. Because many marine mammal species have natural identifying features (dorsal fin shape, fluke colour patterns, body scars, see Fig. 6.1), it is not always necessary to apply a 'mark' in order to recognize unique animals. For that reason, we use the general term 'capture–recapture' (CR) here. Further, the term 'capture' (or 'recapture') connotes any occasion when an individual is identified (or re-identified), which may involve actual physical capture, visually reading an applied tag from a distance, or simply photographing an animal and subsequently matching the photograph to a known identity.

The basics

To demonstrate the concept of CR methods, consider the simplest possible case. We capture, tag, and release 100 seals. A year later we come back, diligently scan the beach and see 35 of our tagged animals. Thus, we conclude the 1-year survival rate was 35% ($35/100 = 0.35$). The problem with this simplistic approach is that, try as we might, we are usually unable to find all the seals that have survived. That is, the *probability of capture* is less than 1, and we have missed some seals that were alive. The key is to estimate how many seals were alive but *not* seen. The objective of CJS analysis, then, is fundamentally to estimate both the probability of survival and probability of capture, given that an animal is alive.

Now let's expand our simple example. We tag our 100 seals and subsequently conduct re-sighting effort at annual intervals. For each of our animals we will compile an 'encounter' or 'capture' history, which is a row of 1s and 0s, indicating whether or not the individual was seen each year. For example:

1101

represents an animal marked in the first year, seen in the second and fourth year, but not seen in the third year. What probabilities underlie this particular encounter history? First, we introduce some notation. It is conventional to denote survival from time i to $i+1$ as Φ_i, and the probability of capture at time i as p_i. In our example, we released our newly tagged seal in the first year of our study, which is indicated by the first '1' above. In the second year, we re-sighted our seal, an event with a probability equal to the product of the chances of surviving and being seen given one is alive: $\Phi_1 \cdot p_2$. On the third occasion, our seal was not seen, which might mean that it died (probability $1 - \Phi_2$, the probability of *not* surviving from year 2 to 3). However, because we saw this wily character in the fourth year, we know he was alive and missed in the third year (probability $\Phi_2 \cdot (1 - p_3)$). Seeing the animal in the fourth year had a probability of $\Phi_3 \cdot p_4$. Finally, the probability of this entire encounter history is simply the product of all of its components:

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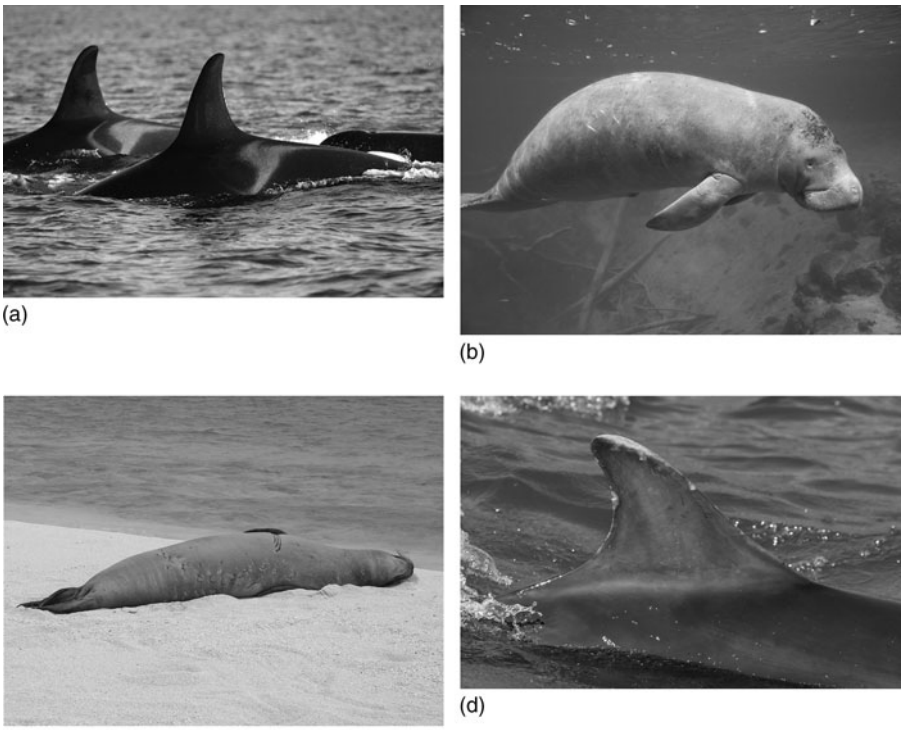

Fig. 6.1 Examples of marine mammals that have been the subject of studies in to vital rates using natural markings and photo-identification. (a) An approximately 35-year-old southern resident killer whale female and her adult daughter in Washington State (photo courtesy of Kim Parsons, Center for Whale Research, Washington, USA). (b) A sub-adult male West Indian manatee approximately 5 years old in Crystal River, Florida (photo courtesy of US Geological Survey, Sirenia Project). (c) A 4-year-old female Hawaiian monk seal at Kure Atoll with scars from a large shark attack (photo courtesy of Jessica Lopez, US National Marine Fisheries Service). (d) A 10-year-old female bottlenose dolphin in the Moray Firth, Scotland, who has been followed over her lifetime by using natural marking for identification. She has also been shown to give birth to two offspring. (Photo courtesy of Barbara Cheney, University of Aberdeen.)

$$\Phi_1 \; p_2 \; \Phi_2 \cdot (1 - p_3)\Phi_3 \cdot p_4$$

In our four-year study, there are eight distinct possible encounter histories. They are:

1111
1110
1101

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1100
1011
1010
1001
1000

CJS survival estimation employs a statistical method called 'maximum likelihood estimation' (MLE). In our example, we would compile the encounter histories for all 100 study subjects and then use MLE to identify which combination of parameter values for the various Φ's and p's is most likely to produce the observed encounter histories. MLE parameter estimates as well as their variances and covariances can be readily computed by a number of computer programs. Program MARK[2] currently implements the broadest range of CR models within a single software package.

Assumptions

The CJS model has a number of associated assumptions. First, it is important to recognize that to estimate true survival, we must assume that our animals do not emigrate from the study area. If they do, then emigrants will appear to have died. Thus, mortality and emigration are confounded and indistinguishable. If emigration cannot be ruled out (or estimated, see p. 135) then in fact CJS models estimate what is termed *apparent survival*, which is the product of the probability of surviving and remaining in the study area.

In addition, the CJS model involves the following assumptions:

- each marked animal has the same probability of capture (p_i) at time i
- each marked animal has the same probability of surviving (Φ_i) from time i to $i+1$,
- marks are permanent and correctly recorded upon capture, and
- all samples are obtained instantaneously, and releases occur immediately.

As with most statistical models, adhering to underlying assumptions is a challenge in practice. There are, however, ways to minimize departure from these assumptions through model structure. The standard CJS model incorporates time-dependent survival and capture probabilities, but can be readily expanded to accommodate a universe of possibilities. The first two assumptions above will be violated if, for example, survival varies with age (as it does in long-lived species), by sex, location, or any number of other factors. Likewise, some animals may be more readily 'catchable' than others because of their age, reproductive status, etc. To account for these differences, age-dependent models can be constructed and

[2] http://www.warnercnr.colostate.edu/~gwhite/mark/mark.htm (see http://www.phidot.org/software/mark/rmark for a handy interface to Program MARK through the free software package R).

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distinct parameters fit for defined groups (i.e. sex, location). In this manner, the first two assumptions are relaxed such that capture and survival probabilities need only be homogenous *within* groups. Within the CJS framework, individual covariates (e.g. body mass) that might influence the probabilities of survival and capture can also be incorporated into model structure. Not only do these modifications of model structure help comply with assumptions, when coupled with model selection (see Section 6.3.4) they provide a mechanism for testing alternative hypotheses about basic biological and ecological relationships.

The third assumption, that marks are permanent and correctly recorded can be another challenge. Mark loss leads to negatively biased survival estimates as animals that lose their identifying marks, like emigrants, appear to have died. In marine mammal photographic identification studies, mark impermanence may involve such things as scars that fade or dorsal fin marks that are obscured by the acquisition of additional marks. Properly applied hot-iron brands are permanent marks that have been successfully used on pinnipeds, but some brands can also become difficult to read in inclement weather conditions and as animals age (see Chapter 2). These issues can be minimized through study design and accounted for quantitatively. Redundancy of marks (e.g. double-tagging, use of implanted passive integrative transponder (PIT) tags), helps ensure that identities are not lost. Redundant marking also facilitates estimating tag loss rates which can be used to adjust survival rates accordingly (Cameron and Siniff 2004). Periodic re-tagging or regularly updating photographic identification databases will help prevent loss of identities.

The final assumption, that samples are collected instantaneously, is of course not possible. In practice, this means that the duration of capture or re-sight effort should be small relative to the interval for which survival is estimated.

6.3.3 Study design considerations

Studies of marine mammal survival require a minimum of two years just to obtain simple return rates uncorrected for sighting probability. In general, it must be anticipated that it will take several years to obtain reliable estimates. At the outset, then, it is especially important to think carefully about study design to maximize the likelihood of success. Indeed in some cases, consideration of the potential errors and biases, together with determination of the accuracy, or precision, required to address the objectives of the study, might show that a study is not feasible. The precision of estimates increases with both the number of animals marked and the probability of recapture. This makes intuitive sense. Let us imagine that 50 animals are marked and the subsequent recapture probability is only 1%. That would mean that on average only one animal would be encountered every other year and the survival estimates would be very uncertain. With most marine mammals, the investigator will likely have some idea about how many animals can be 'marked' and some sense of recapture probability. In Program MARK, one can simulate datasets with specified sample sizes, survival rates, and capture probabilities. Then

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models can be fitted to those data and evaluated as to whether the resulting precision is acceptable. We strongly recommend this.

Adherence to CR assumptions can also be improved through proper study design. Heterogeneity of capture probabilities results from both the intrinsic properties of individual animals (e.g. behaviour or distinctiveness of marks) and from field protocols. If animals do not randomly mix within the study area and, due to convenience or proximity, some parts of the area are more thoroughly searched than others, the result will be that some animals will be more or less likely to be encountered. Distributing re-sighting effort equally throughout the study area will ameliorate this source of heterogeneity. A related concept has to do with variable recognizability of marks, something that is common with photographic identification of cetaceans. Imagine a cetacean study that relies upon dorsal fin marks to identify individuals. An animal with the top half of its fin missing is far more conspicuous than one with small subtle nicks. If both appear in a large group near the study vessel, the lop fin is much more likely to be detected. However, it is possible to equalize these animals sighting probabilities by systematically attempting to photograph each animal in the group rather than allowing undue attention to be focused on the conspicuous marks. This concept can be applied broadly to any situation where some animals are easier to 'recapture' than others.

Finally, individual capture probabilities can be made more homogenous by pooling all sightings over an extended field season. To establish that an animal has survived since the previous year, it need only be seen once in the current year. Animals that have differing probabilities of detection on any given survey day will have more similar probabilities of being seen *at least once* during multiple survey days. To illustrate, consider two tagged seals that spend 30% and 15% of their time on the beach and assume that researchers can detect all seals on land during a survey. Clearly, if only one survey is done, then the first seal is twice as likely to be seen. However, after 10 survey days, the first animal has a 97% chance and the second an 80% chance of being detected at least once—far more comparable capture probabilities. A potential problem with extending survey periods too long is that we begin to run up against the assumption that all samples are obtained simultaneously. The result is that when the field season expands, the actual interval over which we are observing survival begins to vary more among animals. That is, an animal seen late in the survey period one year and early the next has been observed to survive a shorter interval than one seen early one year and late the next. A trade-off, then, is to make the survey period long enough to achieve acceptably high (and as homogenous as possible) capture probabilities, but still relatively short compared to the time between survey periods (usually one year).

6.3.4 Model selection

When analysing CR data, it is customary to fit a variety of models with different parameterization and then select the 'best' model. Model selection can help to

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obtain the most reliable survival estimates and is especially useful for testing hypotheses about ecological factors related to survival. For example, if we want to know whether survival (or capture probability) differs according to sex, colony, or any other factor, we need only build alternative models which fit separate survival (or capture) parameters for each level of the factors of interest and compare them to a 'reduced' model where distinct groups all have the same parameters.

In CR survival analysis, nested models can be compared using the likelihood ratio test (LRT, Lebreton *et al.* 1992). More recently, Akaike's information criterion (AIC) has become a preferred tool for model selection. The theoretical basis and various properties of AIC are beyond the scope of this chapter; however there are several excellent references available (Anderson and Burnham 1999; Anderson *et al.* 2000; Burnham and Anderson 2002). The basic concept of AIC is that it indicates the relative support for different models given the data available. Generally, model fit improves with the number of parameters included, but adding parameters tends to degrade precision. AIC balances the trade-off between model fit and the number of parameters to indicate the most parsimonious model. Note that AIC indicates *relative* support. That is, it does not tell you whether any of your models are good or true; rather it tells you the degree of support *among* the models you have fitted. If you have neglected to include a highly significant parameter in all your models, you will not have anything close to the best possible model, but AIC will still tell you which of your (lousy) models is best.

One advantage of AIC is that, unlike the LRT, non-nested models can be compared as long as the underlying likelihood function is the same. Another feature of AIC is that it frees the analyst from the constraints of arbitrary *p-values*. With traditional methods, we compare two models and declare a winner only if we are 95% certain ($p < 0.05$) that one of the models is superior. Sometimes the critical value is set at 0.01 or 0.10, but in every case this criterion for significance is quite arbitrary. In contrast, AIC provides a measure of relative support with no strict critical value. *Akaike weights* normalize the relative support for each of a set of models based upon their AIC values. The sum of all Akaike weights is 1, and they are denoted as w_i, with i indexing the various models. For example, if you have fitted six different models and there is a clear frontrunner, then the w_i for the best model might be something like 0.92, the next best model might only have a w_i of 0.04, and so on. In cases where several models have similar AIC values, choosing between these is unnecessary. Instead, model averaging can be used, which essentially means obtaining a parameter estimate (and variance) by taking the average value from all the fitted models, weighted by their Akaike weights. By doing so, the uncertainty inherent in model selection is reflected in the model averaged parameter estimates.

The preceding has provided a basic grounding in CJS models for estimating survival in marine mammals. While CJS approaches are most common and can be applied in a wide variety of scenarios, additional methods exist which can provide estimates of parameters other than survival or are better suited to certain types of data.

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6.3.5 Multi-state models

Consider you are studying a species which occasionally moves among distinct habitats. Perhaps survival and capture probabilities vary depending upon where the animals are, either because of varying risks (e.g. predators), differences in behaviour, or differences in accessibility of the sites to you, the researcher. You might consider analysing survival rates using a CJS model and stratifying the individuals by location. The problem with that approach is that a single individual may visit more than one location over time, so that each animal cannot be assigned to a stratum. Multi-state, or multi-strata models, as they are also known, are designed to deal with situations where individuals can move between states which have distinct associated capture and survival probabilities (Brownie *et al.* 1993; Schwarz *et al.* 1993; Nichols and Kendall 1995; Lebreton and Pradel 2002). With this approach, we estimate not only those parameters, but also probabilities of transition between strata at each time step. These transition probabilities are also referred to as movement parameters as they may entail physical movement as in the example above. Another application of multi-state models is for species that do not breed every year, which is often the case among marine mammals. In this case, the strata refer to conditions (i.e. breeder versus non-breeder) rather than a location. Beauplet *et al.* (2006) used multi-state models to evaluate survival rates of subantarctic fur seals (*Arctocephalus tropicalis*), as well as the survival costs of breeding.

6.3.6 Recoveries from dead animals

Until now, we have considered only capture–recapture studies where the observations involve live encounters with animals that had previously been marked. There may be situations whereby marked animals are recovered dead. In these cases, rather than estimating survival and capture probabilities, the parameters of interest are apparent survival and probability of 'recovery'. The latter conflates three probabilities—that of dying, being recovered, and being reported. The original approaches to dealing with these type of data are Brownie models (Brownie *et al.* 1985), and have typically been applied to situations such as ringed game birds recovered during hunts. However, it is not required that the animals die as a result of a hunt (Seber 1970). Analogous situations may occur when marked marine mammals are harvested for food, caught incidentally in fisheries, or found stranded.

In some cases, both live re-sightings and dead recoveries of marked animals may be available to estimate survival. Burnham (1993) and Barker (1997) developed methods for jointly analysing both types of data in a single modelling approach. Under some circumstances, it may be possible to separate true from apparent mortality by using both live encounters and dead recoveries. Dead recoveries and joint live encounter/dead recovery studies are rarely conducted in marine mammals. However, Hall *et al.* (2001, 2002) demonstrate the utility of combining a variety of data sources, including directed research surveys, re-sightings by the public, and opportunistically reported deaths of marked grey seals.

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6.3.7 Robust design

The most complex and, potentially, the most informative CR model is that of Pollock's robust design (Pollock 1982; Kendall *et al.* 1995, 1997). It involves clustered recapture occasions separated by a longer interval. For example, let's say each year we conduct a re-sighting effort five times within a two-week period. Thus, we will have clusters of five capture occasions separated by a year. The key to the classic robust design is that during the two-week sampling period we assume population closure, meaning there are no births, deaths, or emigration or immigration (except under certain restrictive criteria). In contrast, during the year between periods of clustered re-sight effort, the population is open. The power of this approach is that it allows the estimation of a suite of important population parameters, including apparent survival, abundance, and temporary emigration and immigration. If additional data (dead recoveries) are available, true survival and permanent emigration may also be disentangled in some cases.

Finally, the open robust design (Schwarz and Stobo 1997; Kendall and Bjorkland 2001) can be applied when immigration/emigration occur throughout the periods of clustered sampling. This is the case, for example, in pinniped colonies where individuals arrive and depart from the colony during a breeding season. Schwarz and Stobo (1997) used grey seal data to demonstrate the method of open robust design.

6.4 Population models

Here, we discuss population models as mathematical expressions of population size as a function of population parameters and time. In other words, a population model is a mathematical formula (or a set of formulae) that describes the relationship between the abundance (or density), vital rates, and other factors that affect abundance. Although there are infinite possibilities for mathematically modelling change in population size over time, models can be categorized as either deterministic or stochastic. In deterministic models, all parameters (e.g. survival and fecundity) are constants, whereas stochastic models treat at least some parameters as random variables. Often, stochastic models draw parameter values from probability distributions, which in turn have been derived from other studies (for example, of fecundity and survival as described above).

Models can be used for parameter estimation or projection. In parameter estimation, a model (or series of models) is fitted to data and the parameter of interest, e.g. population growth rate, is estimated. When a model is used for population projection, model parameters must be estimated or conjectured. Ideally, uncertainty in parameter estimates should be incorporated into the projection process so that results reflect the real uncertainty of possible future outcomes.

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Although brand new models may be created, many are already available in textbooks and the primary literature (e.g. see Caswell 2001; Williams *et al.* 2001; Clark, J.S. 2007). Rather than reviewing a wide range of models for marine mammal populations, we will discuss key concepts, modelling frameworks, and relevant examples. Models are only as good as the data that go into them. Therefore, we also encourage early consideration of the data needs to support a planned model. If the required data are unlikely to be obtained, consider employing a simpler model structure with lesser data demands.

6.4.1 Exponential and geometric models

One of the first decisions when creating a model is whether or not to treat time as a continuous variable. If time is continuous, all processes are modelled with *differential equations*, whereas *difference equations* are used when time is treated as a non-continuous (or discrete) variable. Although differential equations appear more complex than difference equations, the former involve fewer assumptions about birth and death processes because all vital rates are defined as instantaneous. In contrast, with difference equations in discrete models, we must assume that all births and deaths occur at one point between censuses (Caswell 2001).

The most basic concept of population modelling starts with the following equations:

$$\frac{1}{N}\frac{dN}{dt} = r \tag{6.1}$$

and:

$$\frac{N_{t+1}}{N_t} = R + 1 = \boldsymbol{\lambda} \tag{6.2}$$

where r is the intrinsic per capita growth rate, R is the net discrete per capita growth rate, $\boldsymbol{\lambda}$ is the discrete per capita rate of growth, N is the population size, and t is time. Equation 6.1 treats time as continuous (exponential growth), while equation 6.2 represents discrete time steps (geometric growth). Applying a little calculus and rearranging equations 6.1 and 6.2 yields the simple population models:

$$N_t = N_0 e^{rt} \tag{6.3}$$

and:

$$N_t = N_0 \boldsymbol{\lambda}^t \tag{6.4}$$

The per capita intrinsic growth rate (r) is the net change in the number of individuals in the population during a time interval. When the time increment (dt) is small relative to the life span of the species, r is the difference in per capita birth (b)

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and death (d) rates: $r = b - d$. In the discrete time model, we substituted $\lambda = R + 1$. From equations 6.3 and 6.4, we can see that $\lambda = e^r$.

Population growth rate, even without decomposition into birth and death rates, is a valuable indicator of population status. If r is positive (or $\lambda > 1$), the population is increasing, while decreasing populations have a negative r ($\lambda < 1$). Although very simple, these models can be used to determine the rate of population change from simple count data. For example, Brown *et al.* (2005) fitted a variety of population growth models to a 27-year time series of aerial counts of harbour seals at haul-out sites in Oregon. Building on work by Lande and Orzack (1988), various analytical methods have been developed for computing the population growth rates from simple count data or a time series of abundance (Dennis *et al.* 1991; Holmes 2004; Staples *et al.* 2004). Sometimes such time series are the only information available for certain marine mammal populations. We caution that these methods require that time series data are collected in a consistent manner to avoid introducing unknown biases during the estimation process.

6.4.2 Matrix models

In the previous section, we considered population models based on abundance or count data. Although such models are extremely useful for estimating population growth rates, in cases where age-specific survival and fecundity estimates are available, these can be incorporated into a more complex and informative model.

When we account for differences among groups (ages or life stages) there are as many equations as the number of modelled groups, and linear algebra is used to concisely express them. Consequently, these are often called *matrix models* (Leslie 1945; Lefkovitch 1965; Caswell 2001). The scalar components of equation 6.2 are replaced by vectors and matrices:

$$\underline{\mathbf{n}}_{t+1} = \boldsymbol{L}\underline{\mathbf{n}}_{\boldsymbol{t}} \tag{6.5}$$

where $\boldsymbol{n}_t$ is a column vector of the numbers of individuals in each age or stage class at time t, and $\boldsymbol{L}$ is a matrix that specifies birth and survival rates of each age or stage.

As the time subscripts in equation 6.5 suggest, time is treated as a discrete variable in matrix models. The simplest form of the matrix in equation 6.5 is called the *Leslie matrix*, in which only entries for survival rates and per capita fertility are included for each age class. In marine mammals, it is often difficult to estimate these parameters for every age. Consequently, consecutive ages are often combined to form stage classes (*Lefkovitch matrix*, Lefkovitch 1965; Caswell 2001). In these cases, the matrix will contain entries representing the probability of remaining in each stage at each time step. These matrices (Leslie or Lefkovitch) are collectively called projection matrices. Brault and Caswell (1993) used published data (Bigg *et al.* 1990; Olesiuk *et al.* 1990a) to parameterize a stage-based matrix for killer whales (*Orcinus orca*) in the Pacific Northwest. The model stages included yearlings, juveniles, reproductive adults, and post-reproductive adults.

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Matrices have associated scalars and vectors called eigenvalue–eigenvector pairs. Their mathematical definitions are beyond the scope of this chapter, but for now it is sufficient to know that eigenvalues and eigenvectors embody key properties of populations represented by the projection matrix. The dominant eigenvalue, or the largest eigenvalue, is the asymptotic population growth rate (λ) and the population age structure will stabilize in the same proportion as the *dominant right eigenvector* (stable age distribution).

The matrix approach can be extended to model metapopulations and multi-state systems (Caswell 2001). In metapopulation models, multiple locations and movements between locations are modelled along with fecundity and survival. Just as with multi-state capture–recapture models discussed earlier, multi-state population models can address situations where individuals may exist in a variety of states, between which they transition over time. Hunter and Caswell (2005) demonstrate a metapopulation model in which the spatial distribution of a population is explicitly modelled within a matrix approach. They demonstrated that the population growth rate can be computed from a multi-state, metapopulation matrix using linear algebra.

Perturbation analysis is a key aspect of matrix modelling. According to Caswell (2001) 'The results of perturbation analysis are often more interesting, more robust, and more useful than the parameter estimates themselves.' Two common types of perturbation analysis are sensitivity and elasticity analysis. Elasticity provides the proportional response of λ to a proportional perturbation to vital rates (Caswell 1978; de Kroon *et al.* 1986), whereas sensitivity provides information about the magnitude of change in λ with respect to each element of the matrix (i.e. additive perturbation). McMahon *et al.* (2005) used elasticity analysis to show that, in general, juvenile survival rates contributed the most to the change in population growth rates of several southern elephant seal (*Mirounga leonina*) populations.

Despite the apparent complexity of linear algebra notation used in matrix models, they really only differ from the geometric model (equation 6.4) in terms of the type of data they incorporate. Mathematically, equations 6.4 and 6.5 are equivalent. Specifically, equation 6.4 is a special case of equation 6.5, where N_t is a scalar rather than a vector.

6.4.3 Incorporating uncertainty: deterministic versus stochastic models

Matrix models are based upon multiple life-history parameter estimates, the precision of which plays a critical role in the precision of the estimated population growth rate derived from the model. When reporting the estimated population growth rate, for example, it is important to convey the uncertainty around the estimate. Some methods for computing the uncertainty for estimated growth rates include series approximations, bootstrapping (e.g. Brault and Caswell 1993), Monte Carlo analysis (Caswell 2001), and Bayesian approaches (e.g. Goodman 2004; Thomas *et al.* 2005).

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Recall that when birth and death rates in a population model are treated as constants, the model is considered deterministic. In reality, we know full well that vital rates are not constant over time. A stochastic model accounts for variability in vital rates over time; i.e. process error or process variability. Information about the variability of vital rates can be incorporated directly into a stochastic model by representing parameters with probability distributions rather than constants. Ideally, these probability distributions will be based on vital rates and their variability estimated using methods described earlier in this chapter. For example, Runge *et al.* (2004) used a stochastic matrix approach to model a population of Florida manatees (*Trichechus manatus latirostris*). By incorporating uncertainty, the projection of the population based on the stochastic matrix will include a range of possible outcomes, rather than a single outcome based on a deterministic matrix. Stochastic models are more informative than deterministic models for managers and conservationists because a distribution of possible outcomes will provide a more realistic view of future uncertainty.

When developing a stochastic model, demographic stochasticity and environmental variability may be incorporated. The former refers to the random chance variability in demographic outcomes, whereas the latter involves variability in vital rates due to environmental factors. In general, environmental variability tends to have the greatest influence on the population growth rate, especially for moderate to large size populations. Incorporating these uncertainties (especially environmental variability) into population models becomes critical for estimating the probability of persistence or extinction (see Section 6.4.6 on population viability analysis).

6.4.4 Density dependence

The models discussed so far have had no upper limits on population size. In reality, of course, a population cannot grow forever. Several biological and physical factors, such as food and space limitations, aggression, predation, emigration, and disease, can reduce the population growth rate as the population size increases. Effects of density on population growth rates (or vital rates) are called density dependence.

The geometric and exponential models (equations 6.1 and 6.2) can be extended to include carrying capacity (K), which can be thought of as the equilibrium population level. For example, the following two equations are particular forms of continuous and discrete time models, respectively, with carrying capacity:

$$\frac{1}{N}\frac{dN}{dt} = r\left(1 - \frac{N}{K}\right) \tag{6.6}$$

and:

$$N_{t+1} = N_t + RN_t\left[1 - \left(\frac{N_t}{K}\right)^{z}\right] \tag{6.7}$$

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where z defines the rate at which the population reaches the carrying capacity. In density-dependent models, the population growth rate becomes a function of abundance (or density).

Note that for the equations above, the rate of population growth decreases as N increases, finally reaching stability (no growth) when $N_t = K$. Although density dependence is an important factor in the population dynamics of large mammals, including marine mammals (Fowler 1981), it is not always included in models. One reason is that the growth rate of a small recovering population may be affected little by its density. In general, if only short-term projections of a small population are performed using models of long-lived species like marine mammals, effects of density dependence may be negligible. Another reason for omitting density dependence in population models is lack of information on how density actually affects vital rates in the species and population of interest. While general patterns may be anticipated (Fowler 1981), obtaining information about the precise functional form of density dependence requires observing vital rates or population growth rates of a population over a wide range of abundances from zero to K, or perhaps by comparing observed rates among multiple populations at different population levels. Alternatively, information from similar species can be used to define some reasonable functional form for density dependence. We caution that model results can be highly sensitive to how density dependence is implemented, especially for long-term projections.

Density dependence usually refers to reduced population growth as abundance increases. There also exists inverse density dependence at low population sizes, often referred to as an Allee effect (Odum and Allee 1954), which describes diminished per capita population growth rates in populations at very low abundance. Allee effects are often associated with difficulty finding mates at low population levels or a breakdown of breeding, foraging, or other key functions that depend upon some minimum density in social species (such as some marine mammals). Dennis (2002) has shown that Allee effects are important to consider for decreasing populations. When modelling a small and declining population, including Allee effects may avoid overly optimistic projections of population growth that could result from including only compensatory density dependence.

6.4.5 Individual-based models (IBM)

We began with modelling an entire population as a group, then introduced matrix models that account for groups (ages, stages) of individuals with distinct associated vital rates. However, the fundamental unit of a population is an individual. The IBM approach explicitly deals with the variability at the individual level. Rather than using a population growth model (e.g. equations 6.1 and 6.2), each individual is modelled with birth/death processes and other factors that affect its survival and reproduction. In essence, the fate of each individual in a population is tracked through time. Such a modelling approach allows incorporation of more detailed

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information about the population than when a population is treated as a whole. For example, a genetic component may be added to a model so that the survival rate of an individual is based on its fitness, which may be a function of its parents' genotypes. IBMs involve intense simulations and demand much computer processing capacity, as each individual has its own probability functions for demographic rates. Further, and more importantly, a considerable amount of data is required about survival, reproduction, and all other parameters to support an individual-based model (cf. Grimm and Railsback 2005).

Hall *et al.* (2006a) used the IBM approach to examine the effects of PCBs (polychlorinated biphenyls) on the potential population growth rate of bottlenose dolphins (*Tursiops truncatus*) in Sarasota Bay, Florida, perhaps the best studied cetacean population in the world (Scott and Chivers 1999; Scott *et al.* 1990a, b; Wells and Scott 1990). They simulated the process of accumulation and depuration of PCB in the dolphins and modelled first-year calf survival as a function of maternal PCB blubber concentration. They found that they had insufficiently precise data to make definitive conclusions about how PCB affects the population, even though they tentatively concluded that the current environmental levels of PCB were depressing the population's growth rate. The observed population growth rate, however, was found to be outside the 95% confidence interval estimated from the model simulations. This example shows that even for a well-studied population, data may be insufficient to satisfactorily parameterize an IBM. Currently, we do not believe that any marine mammal population has sufficient data to support a meaningful IBM.

6.4.6 Population viability analysis (PVA)

The aim of PVA as originally developed was to evaluate population viability measured in terms of extinction risk within some time horizon (e.g. 100 years). Recently, it has been argued that such estimates of extinction risk may be not accurate or credible (Ludwig 1999; Ellner *et al.* 2002). The reliability of a PVA depends on having information on the population itself (e.g. abundance, vital rates, age–sex structure) as well as such factors as environmental variability, demographic variability, density dependence (including Allee effects), catastrophic events, and anything else that may affect the population. If such factors are not correctly characterized in the model, or uncertainty is inadequately expressed, then PVA results will likely be misleading.

Even with the difficulty in obtaining all this information, there is still a role for PVA in marine mammal conservation and management. While we may never achieve a precise and accurate estimate of extinction risk, a PVA can provide a mechanism for organizing and integrating existing relevant data that bears on extinction risk while characterizing associated uncertainty.

Several books have been written on this subject (e.g. Soulé 1987; Beissinger and McCullough 1992; Ballou *et al.* 1995; Fiedler and Kareiva 1998). Morris and Doak

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(2002) discuss a series of considerations for using PVA. These include focusing on uncertainty (e.g. confidence intervals rather than point estimates), considering extinction risk metrics as relative and qualitative rather than absolute, limiting projections into the future to a short period, considering potential effects of overlooked factors, comparing multiple models if possible, and updating a PVA as more data become available. Running sensitivity analyses on a PVA to determine which parameters most influence the results can aid in directing research to fill key data gaps. Finally, we caution that although 'off the shelf' PVA software is available (RAMAS and VORTEX), the output of such programs should be viewed critically and presentation of results should explicitly document model structure, assumptions, and uncertainty about input parameters.

6.4.7 Bayesian approach to modelling

Although discussed separately, we emphasize that the Bayesian approach is not independent of the rest of this chapter. Any model can be analysed with Bayesian statistics, the use of which has been increasing in recent years. A detailed treatment of Bayesian statistics is beyond the scope of this chapter. A brief non-mathematical introduction can be found in Eguchi (2008).

One application of the Bayesian approach to population modelling is state–space modelling. In a state–space modelling approach, the state of the quantity of interest (e.g. abundance over time) is modelled with a mathematical function such as a population growth model. Observed data are treated as samples from the state, where another mathematical function is used to depict the sampling process. For example, Craig and Reynolds (2004) used the first-order Markov process to model the population size of manatees along the Atlantic coast of Florida. They included movement, sighting probabilities, and abundance in the model. Goodman (2004) showed that multiple independent datasets (mark–recapture and stranding surveys) can be used in a model to improve estimates of population parameters, such as abundance and population growth rate. Thomas *et al.* (2005) used a matrix population model and state–space modelling to unify the data collection process and the population growth model. Parameter inference was obtained via the Bayesian approach. They demonstrated the method by modelling the British grey seal metapopulation. Another strength of the Bayesian approach is the ease in which mixed-effects and random-effects models are analysed. Clark (2007) provides an excellent textbook on applied Bayesian modelling for ecological data.

6.4.8 Model fit and selection

Several candidate model formulations may be generated to estimate parameters of interest such as population growth rate. Just as we described with regard to capture–recapture estimation, AIC and its modifications can be used to select the best amongst a suite of models, or to obtain model-averaged parameter estimates where the weight of each model is proportional to its model selection score

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(Burnham and Anderson 2002). R.F. Brown *et al.* (2005) provide an example of this process applied to a harbour seal population. They fitted multiple candidate models (density dependent, density independent, site-specific growth rate, and constant growth rate) to seal counts. This process of model selection and parameter estimation is common regardless of one's affiliation to statistical philosophy. In addition to AIC, there are analogous Bayesian model selection criteria (Draper 1995; O'Hagan 1995; Spiegelhalter *et al.* 2002).

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